BPMN
Business Process Model and Notation

Modeling Language Components

- A modeling language consists of three parts:
  - ***Syntax***
    - Set of modeling elements and rules to combine them
      - BPMN syntax includes activities, events, gateways, sequence flows
  - ***Semantics***
    - Bind syntactical elements with textual descriptions to a precise meaning
      - Behavior of BPMN elements
  - ***Notation***
    - Defines graphical symbols for elements

Business Process Model and Notation (BPMN)

- Object Management Group (OMG) standard
  - http://www.omg.org
  - http://www.omg.org/specs/bpmn
  - version 1.0 (2006)
  - version 1.1 (2007)
  - version 1.2 (2009)
  - version 2.0 (2011)

- Provide a notation to describe Business Process understandable by:
  - BP analysts
  - IT developers
  - BP workers and managers

Why BPMN

- Business people are very comfortable with visualizing Business Processes in a flow-chart format

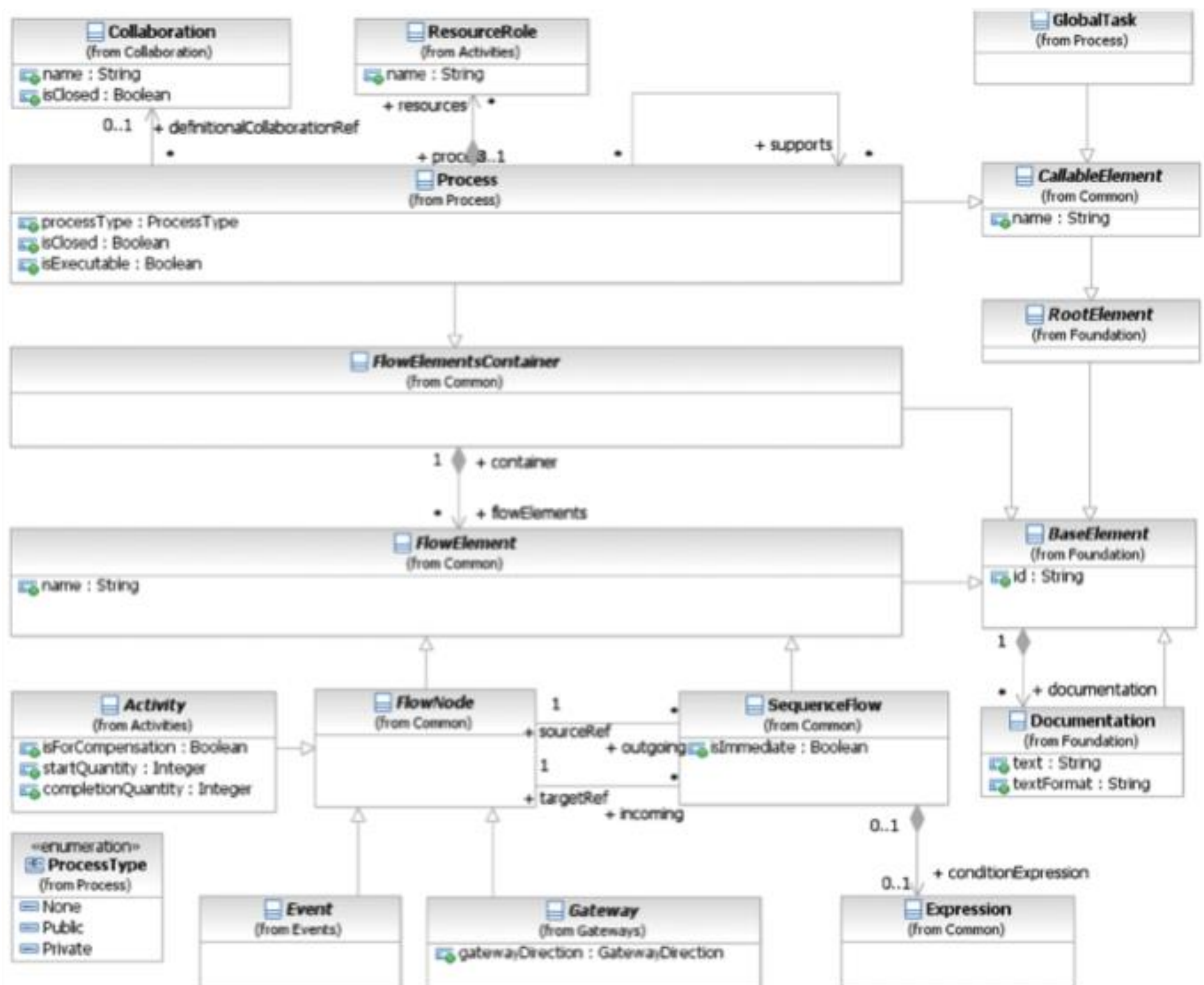
- BPMN ***execution semantics*** is fully formalized

- BPMN 2.0 has a ***formal definition*** (metamodel)
  - Precise definition of the constructs and rules for creating models

BPMN Metamodel

- Metamodeling has the following benefits
  - Formalization of models and entities
  - Formalization of relationship between elements
  - Interoperability
- It is not necessary for the modeler to handle the metamodel
- Is the modeling tool that ensures the model is compliant with metamodel

BPMN Metamodel (partial)



BPMN Semantic

- To describe how the BPMN elements behave, the theoretical concept of ***token*** is used
  - The "***simulated***" behavior of elements can be defined by describing how they interact with a token
  - The token is not part of the BPMN specification
  - BPMN modeling tools are not required to implement any form of token

- The token will traverse the sequence flows and pass through the elements in the process
  - Token traverse sequence flows instantaneously, so there is no time associated with them
  - When arrives at an element, the token may continue instantaneously or can be delayed depending on the element

- We use this notation for tokens 

Sequence Flow

Sequence Flow

- Sequence flow connects model elements showing their ***order of execution***

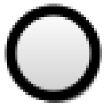
- Each sequence flow has only one source and only one target
  - Model elements can have one or more incoming sequence flow and one or more outgoing sequence flow but usually they have one

- Source and target can be activities, events, gateways

Start Event

- Beginning of the process
  - It has no incoming flows

- A process may have zero, one or more start events
  - If not present, all activities without incoming flows start together
  - If an end event is present, at least one start event is mandatory


End Event

End Event

- Where the flow of the process ends
  - It has no outgoing flows
- A process may have zero, one or more end events
  - If not present, all activities that do not have any outgoing flows mark the end of a process path
    - In this case, the Process ends when all parallel paths have completed
- If a start event is present, at least one end event is mandatory

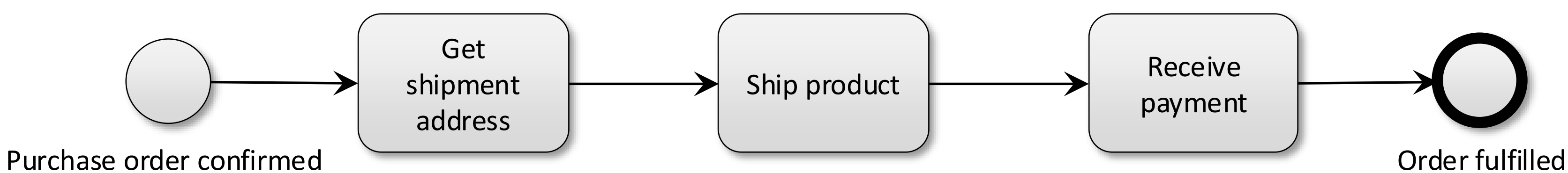
Task

- Unit of work (atomic activity), the job to be performed

- A task is used when the work in the process cannot be broken down to a finer level of detail

Our First BPMN Model

- A simple BPMN model generally has
  1. A start point
  2. An end point
  3. Some activities
  4. Connections between activities



Gateway

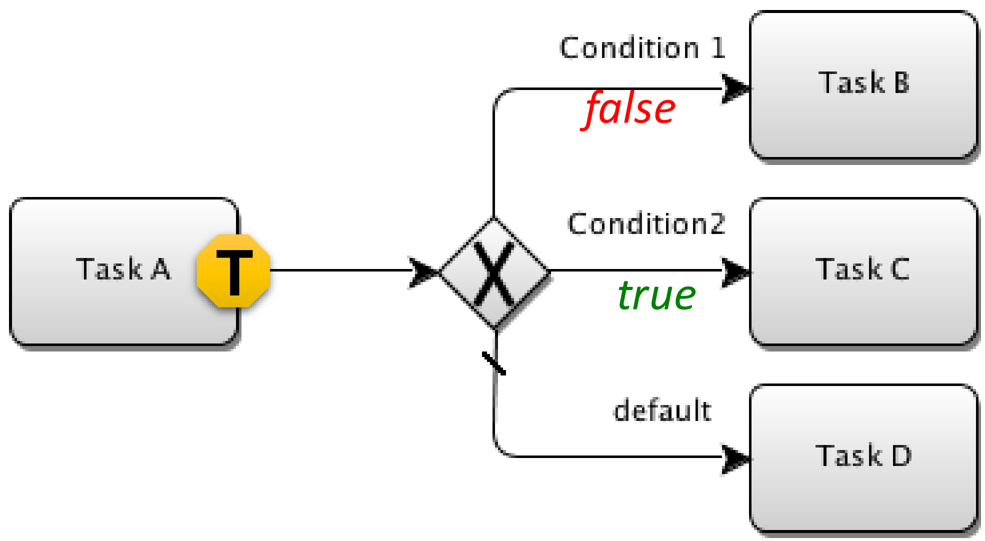
- Used to control how sequence flows interact as they converge and diverge within a process

- ***Split gateway***
  - Point where the process flow diverges
  - Has one incoming sequence flow and multiple outgoing sequence flows

- ***Merging gateway***
  - Point where the process flow converges
  - Has multiple incoming sequence flows and one outgoing sequence flow

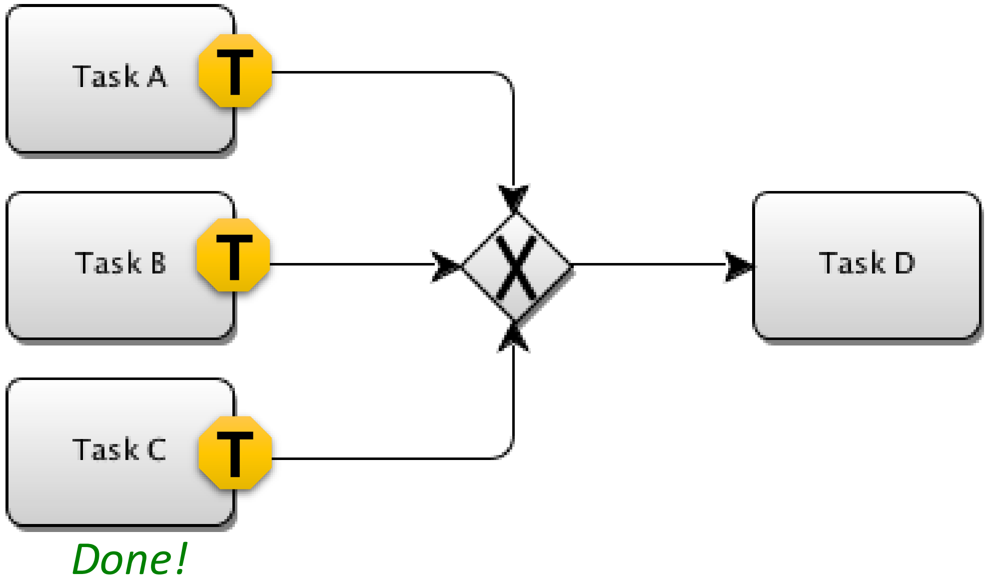
Exclusive Gateway

- ***Diverging Exclusive Gateway***
  - Creates alternative paths within a process flow
  - The decision is considered as a question with a defined set of alternative answers
  - Only one of the paths can be taken (XOR)
  - The default path (if defined) is taken if none of the conditional expressions evaluate to true



Exclusive Gateway

- ***Converging Exclusive Gateway***
  - Used to merge alternative paths
  - Each incoming token is routed to the outgoing sequence flow without synchronization



Exclusive Gateway Example

- Order shipment
  - *When a purchase order is received, check stock availability. If the item is available, confirm order and ship product, otherwise reject order*.

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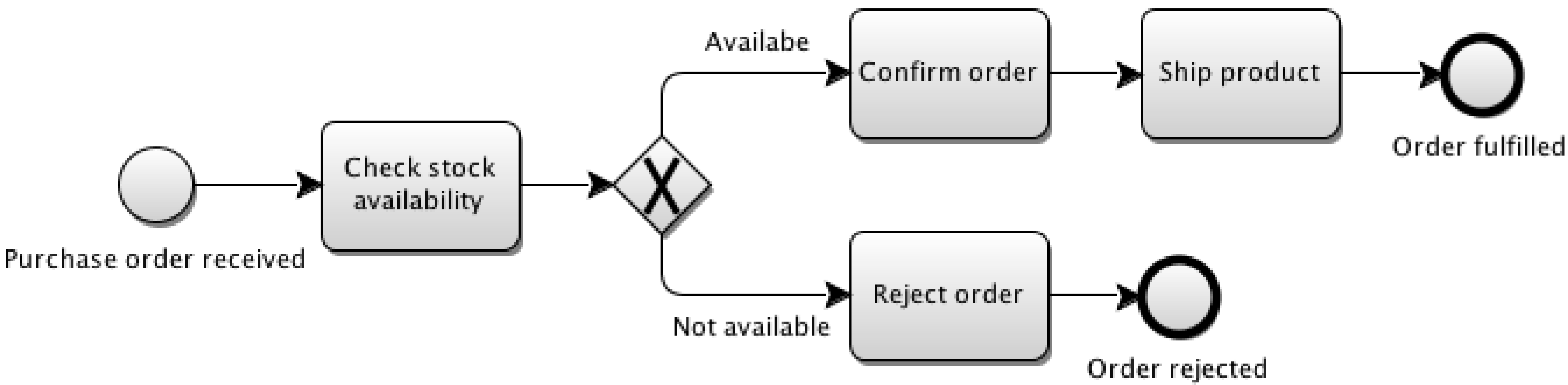
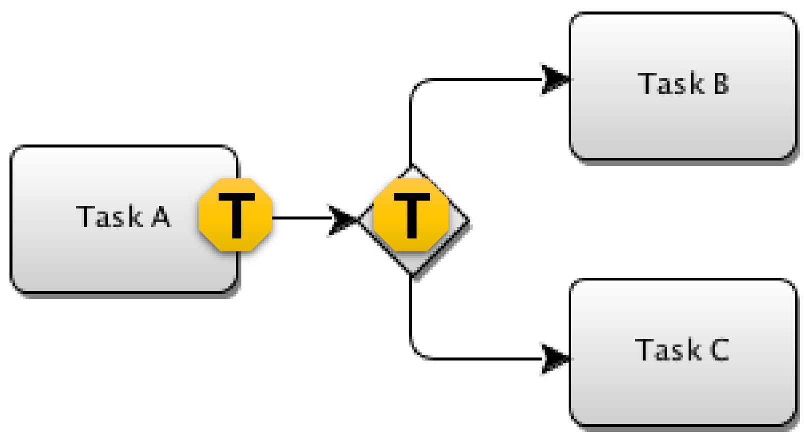


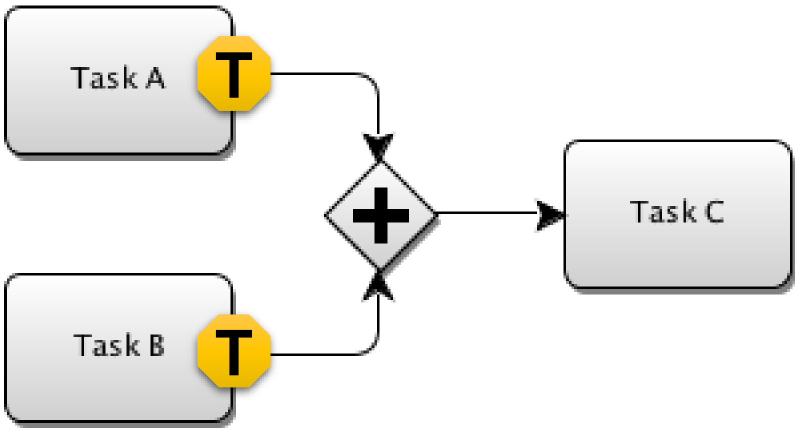
Parallel Gateway

- ***Diverging Parallel Gateway***
  - Creates parallel paths without checking any conditions
  - Each outgoing sequence flow receives a token



Parallel Gateway

- ***Converging Parallel Gateway***
  - Used to synchronize (combine) parallel flows and to create parallel flows
  - Waits for all incoming flows before routing the token to the outgoing flows (AND)



Parallel Gateway Example

- Boarding security check
  - *Having the boarding pass, go to security check for luggage and security screening. When passed both go to boarding*

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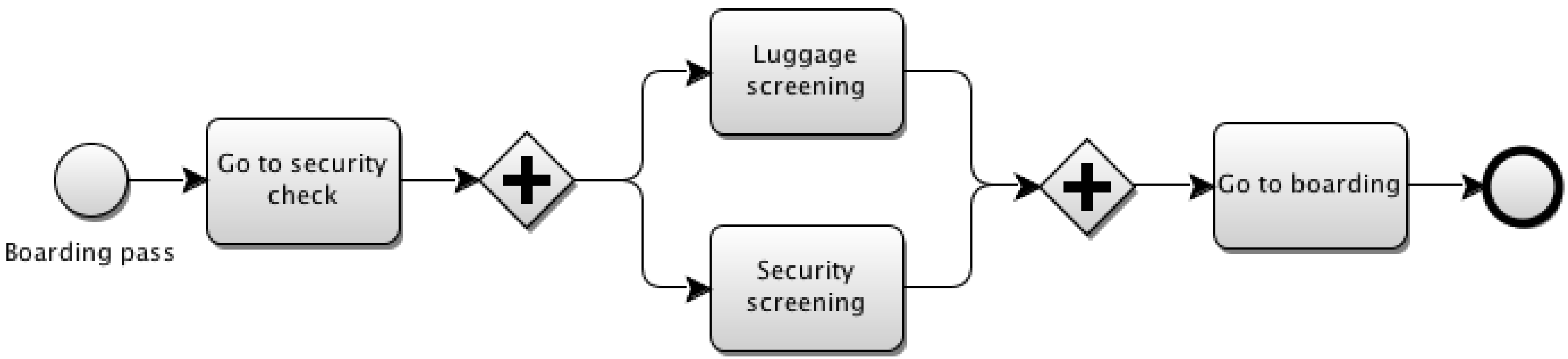
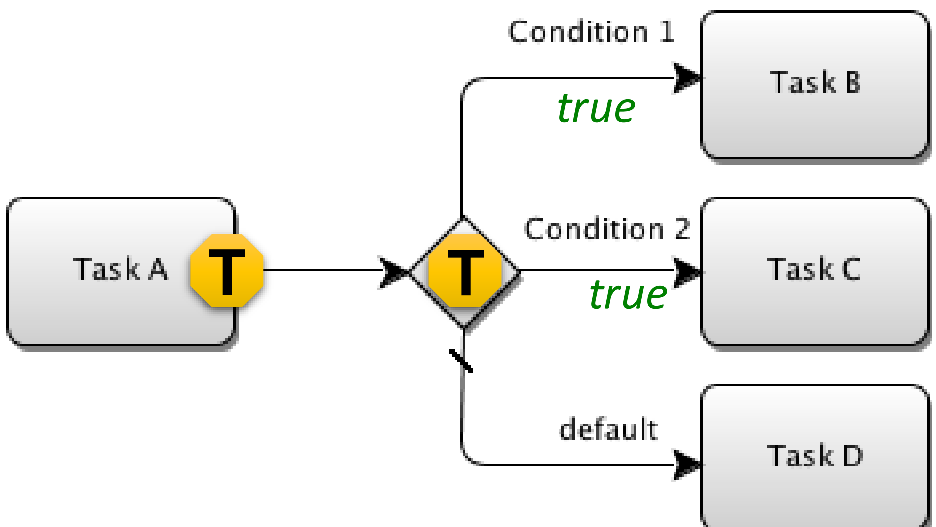


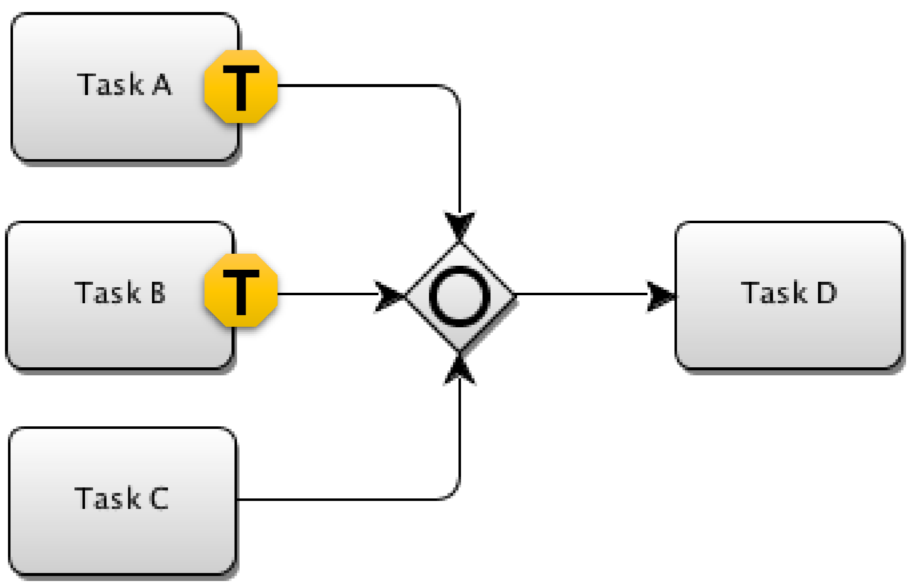
Inclusive Gateway

- ***Diverging Inclusive Gateway***
  - Used to create alternative but also parallel paths within a process flow
  - Unlike the exclusive gateway, all condition expressions are evaluated
    - All Sequence Flows with a true evaluation will be traversed by a token (OR)
    - The default path (if defined) is taken if none of the conditional expressions evaluate to true



Inclusive Gateway

- ***Converging Inclusive Gateway***
  - Used to merge a combination of alternative and parallel paths
  - Token arriving at an inclusive gateway may be synchronized with some other tokens that arrive later at the gateway
    - When all the expected tokens have arrived the token moves to the outgoing sequence flow



Inclusive Gateway Example

- Order decomposition
  - *When an order is received, get product of type A from warehouse A and products of type B from warehouse B.*

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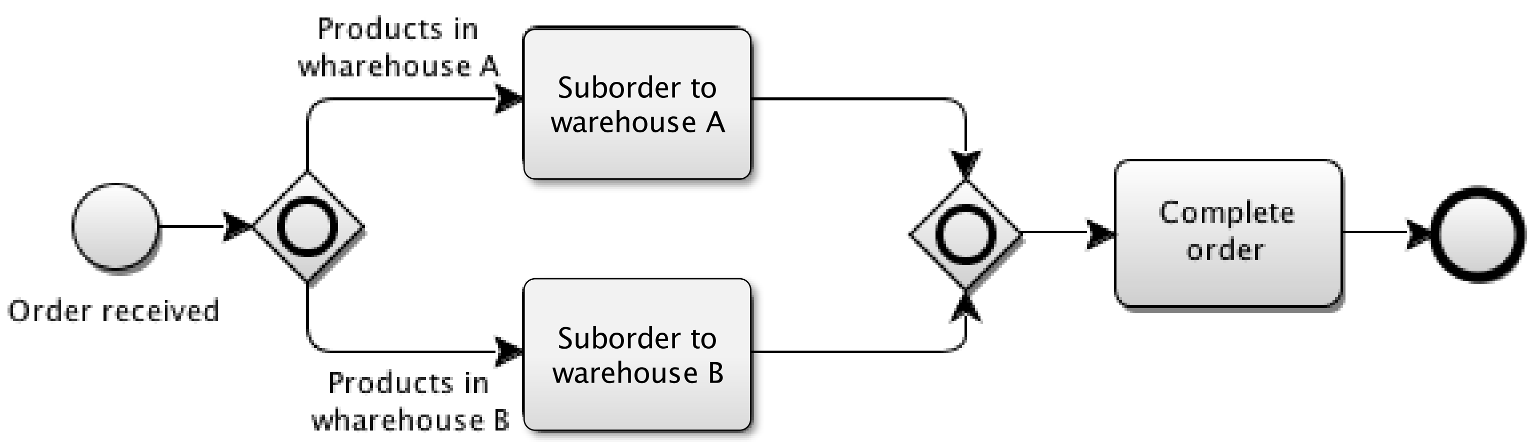


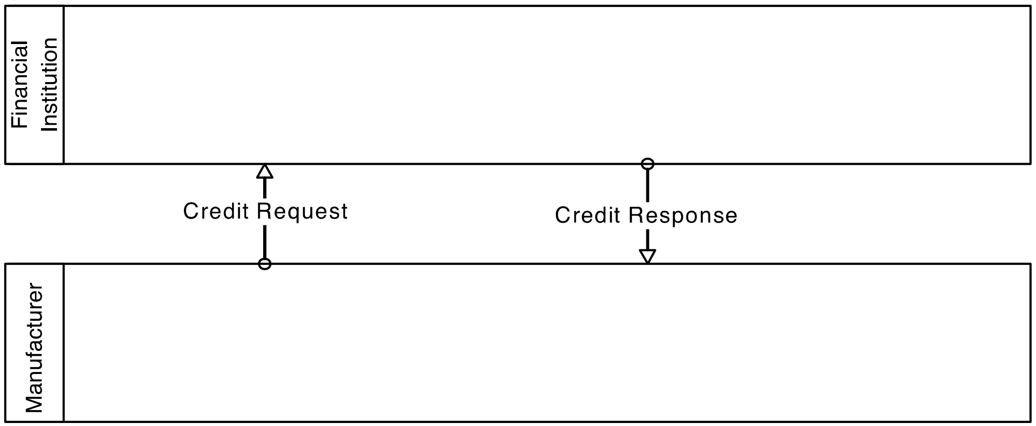
Resource Modeling

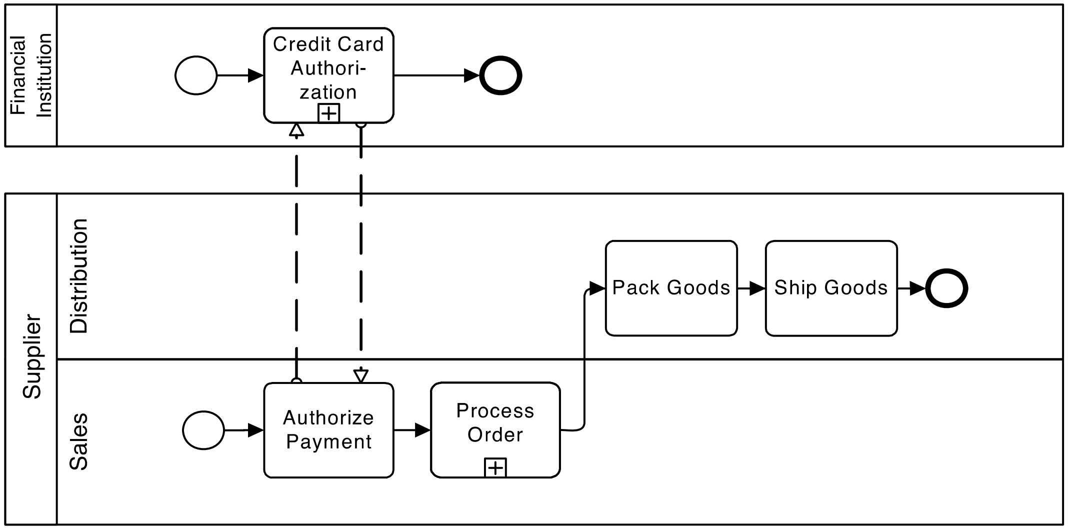
- Organizational resource are mapped in pools and lanes

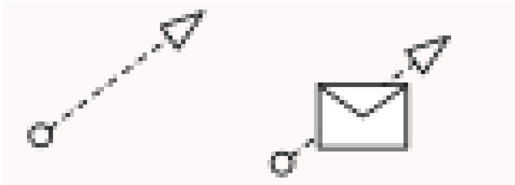
- ***Pools***
  - Independent organizational entities (***do not share*** any common system that allows them to communicate implicitly)
    - Customer is independent from the Supplier

- ***Lanes***
  - Multiple resource classes in the same organization (***share*** common systems)
    - Sales department and marketing Department

Pools and Lanes







Message Flow

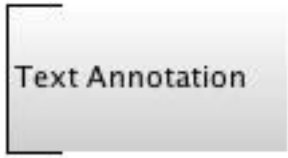
- Used to show the flow of messages between two participants
  - Participants are prepared to send and receive the messages
- Message flow only connects two separate pools
  - Message flow can be attached to pools, activities, or message events
  - Message flow cannot connect two objects within the same pool

Artifacts

- Additional information that is not directly related to the sequence flows or message flows
  - Associations, groups, text annotations

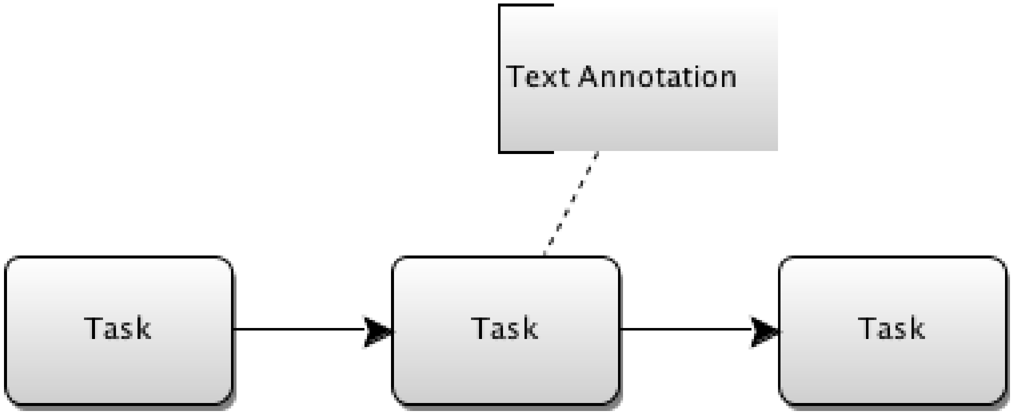
- Association is used to associate artifacts with process elements

Association



Text Annotation

- Provides additional information for the reader of a BPMN diagram
  - Does not affect the flow of the process

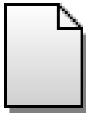


Data Objects

- Represent data and document exchanged in the process
- Can be used to show input and output of activities
- BPMN defines 5 kind of data objects

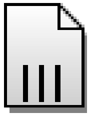
Data Objects

- ***Data object***
  - Information traversing the process (email, letters, documents)



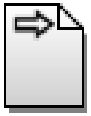
Data Object

- ***Collection data object***
  - Collection of information



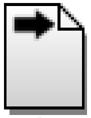
Collection Data Object

- ***Data input***
  - External input necessary to start the activity



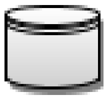
Data Input

- ***Data output***
  - Outcome of the activity or the process



Data Output

- ***Data store***
  - Place where the process can read and write data (database)



Data Store

BPMN Process Type

- ***Orchestration***
- ***Collaborations***
- ***Choreographies***

- BPMN uses the terms collaboration and choreography when modeling the interaction between processes

Orchestration Example

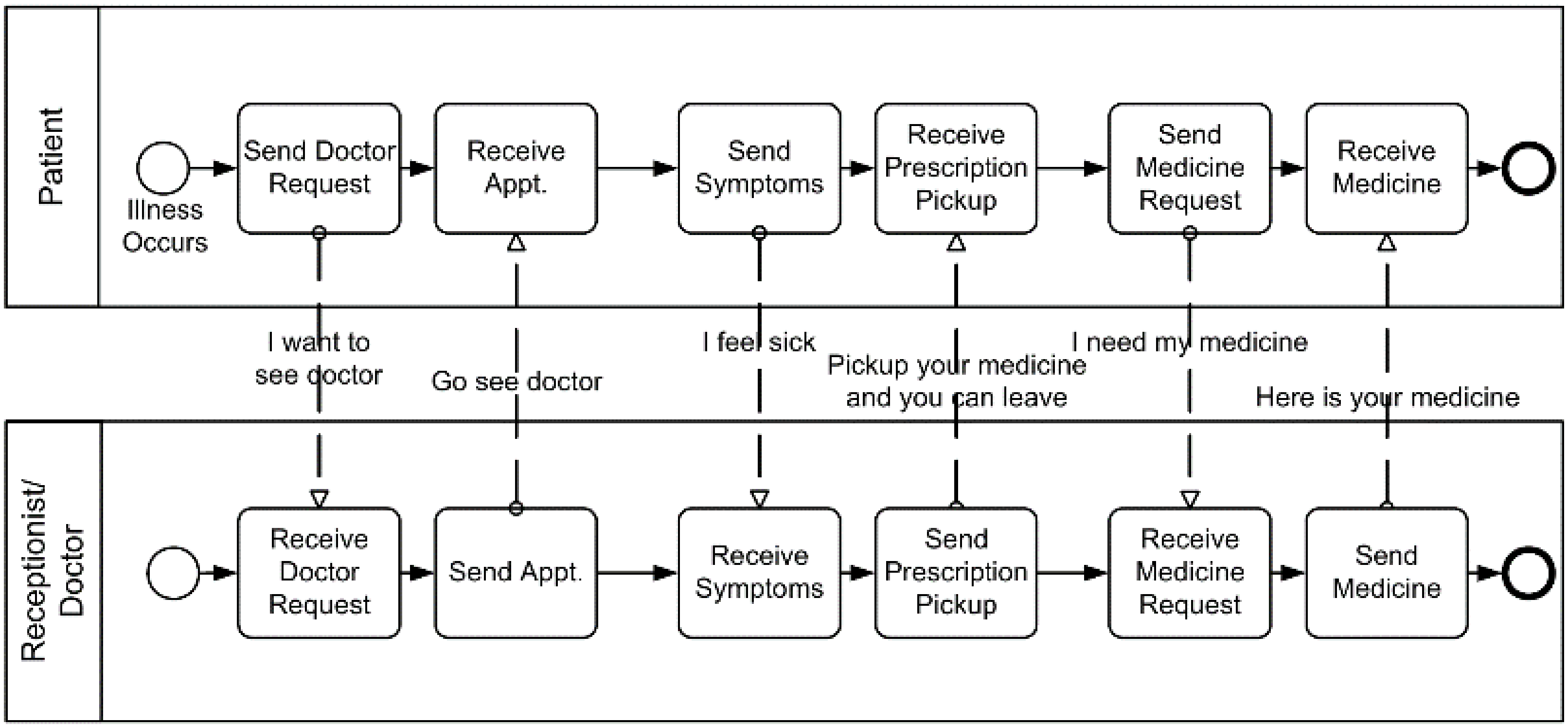
- BP ***internal*** to a specific organization
- Generally called workflow or BPM process

Image from "Business Process Model and Notation (BPMN)", http://www.omg.org/spec/BPMN/2.0

Collaboration

- Collaboration shows ***interactions*** between two or more participants
  - A pool represents one of the participants in the collaboration
- Collaboration can be shown as two or more processes communicating with each other
  - The message exchanged between the participants is shown by a message flow

Collaboration Example



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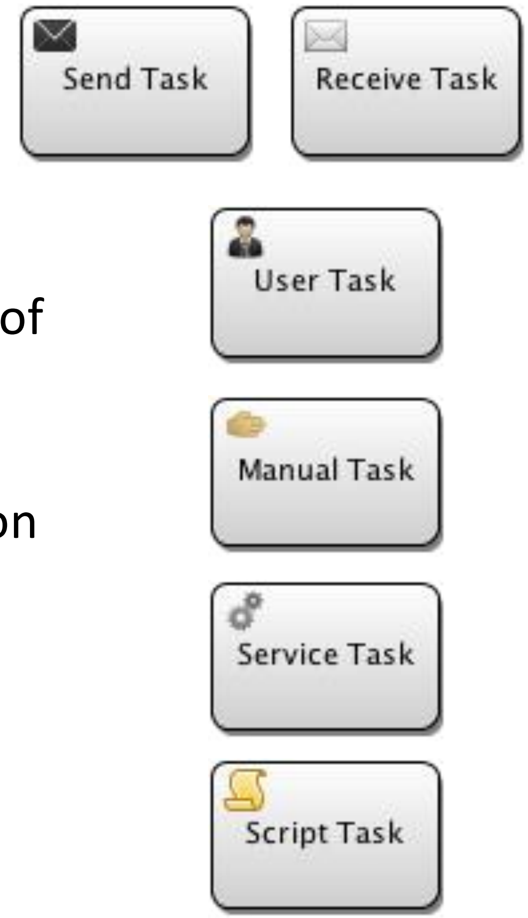
Specialized Task

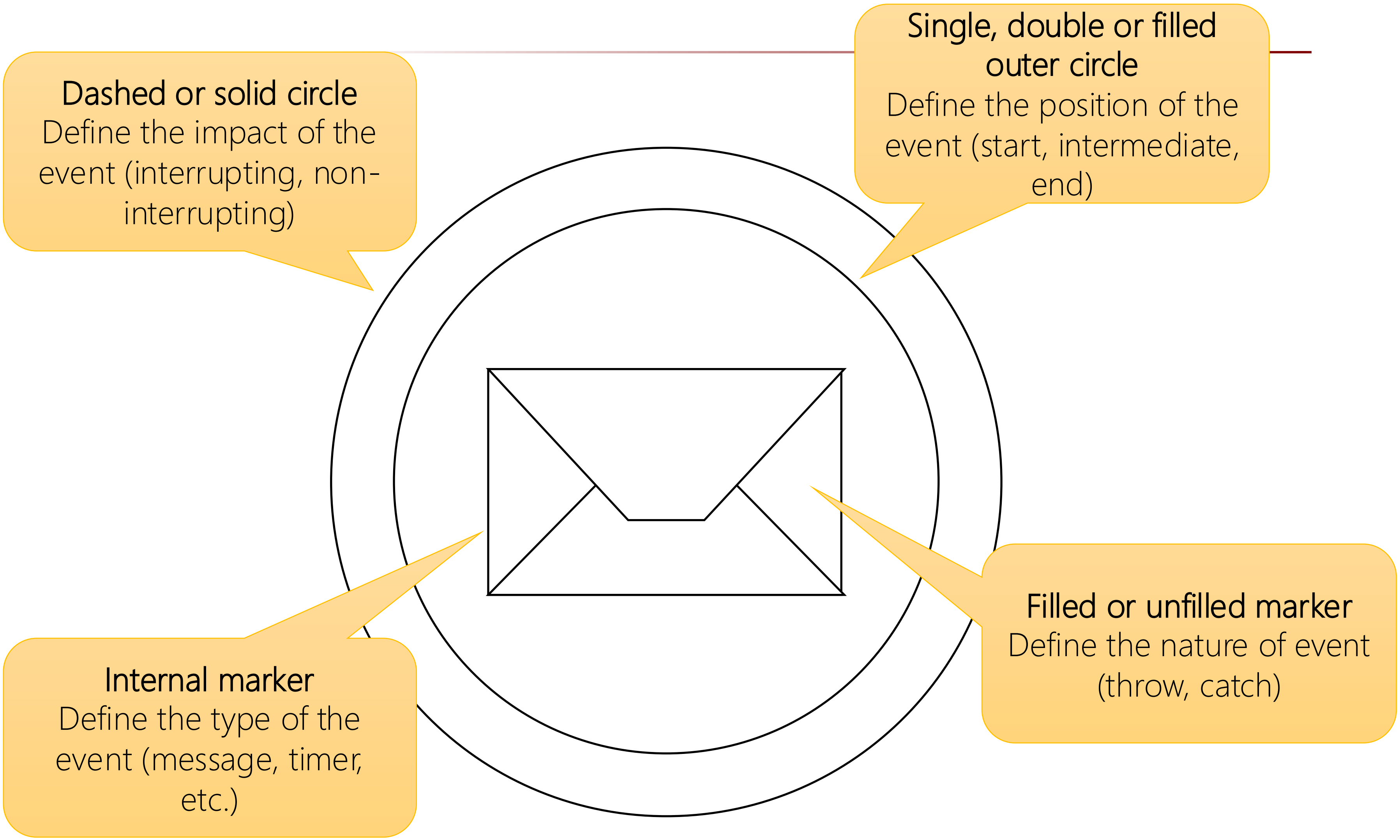
- ***Send/Receive message***
  - Task that send/receive a message to/from an external participant (within the process)
- ***User activity***
  - Task performed by a human with the assistance of a software application
- ***Manual activity***
  - Task performed without the aid of any application
- ***Service calling***
  - Task that uses some sort of external service identified with an URI
- ***Script***
  - Task performed by a business process engine



Event Definition

- The “Event” is the composition of
  - ***Position***
    - Start, Intermediate, End
  - ***Nature***
    - Catch, Throw
  - ***Impact***
    - Interrupting, Non-Interrupting
  - ***Type***
    - Message, Timer, etc.

Event Symbol



Event Position

- Start event
  - initial point of execution flow

- Intermediate event
  - may occur between start and end of the execution flow

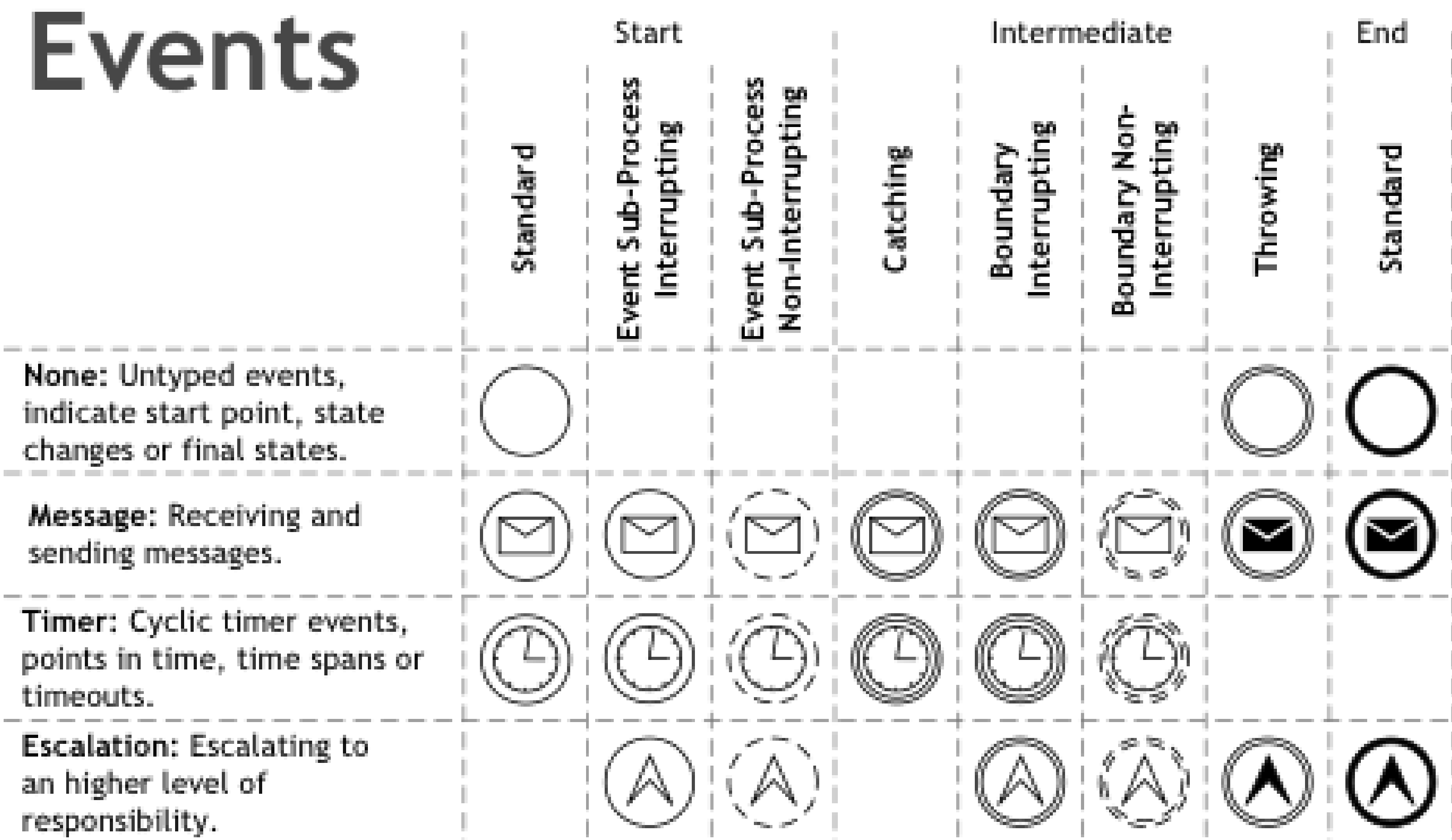
- End event
  - stop of the execution flow

Event Nature

- ***Catch event***
  - All start events and some intermediate events
  - When the event is triggered, the token is generated

- ***Throw event***
  - All end events and some intermediate events are throwing events that may eventually be caught by another event
  - Typically, the trigger carries information from the scope where the throw event occurred into the scope of the catching events

Events Definition



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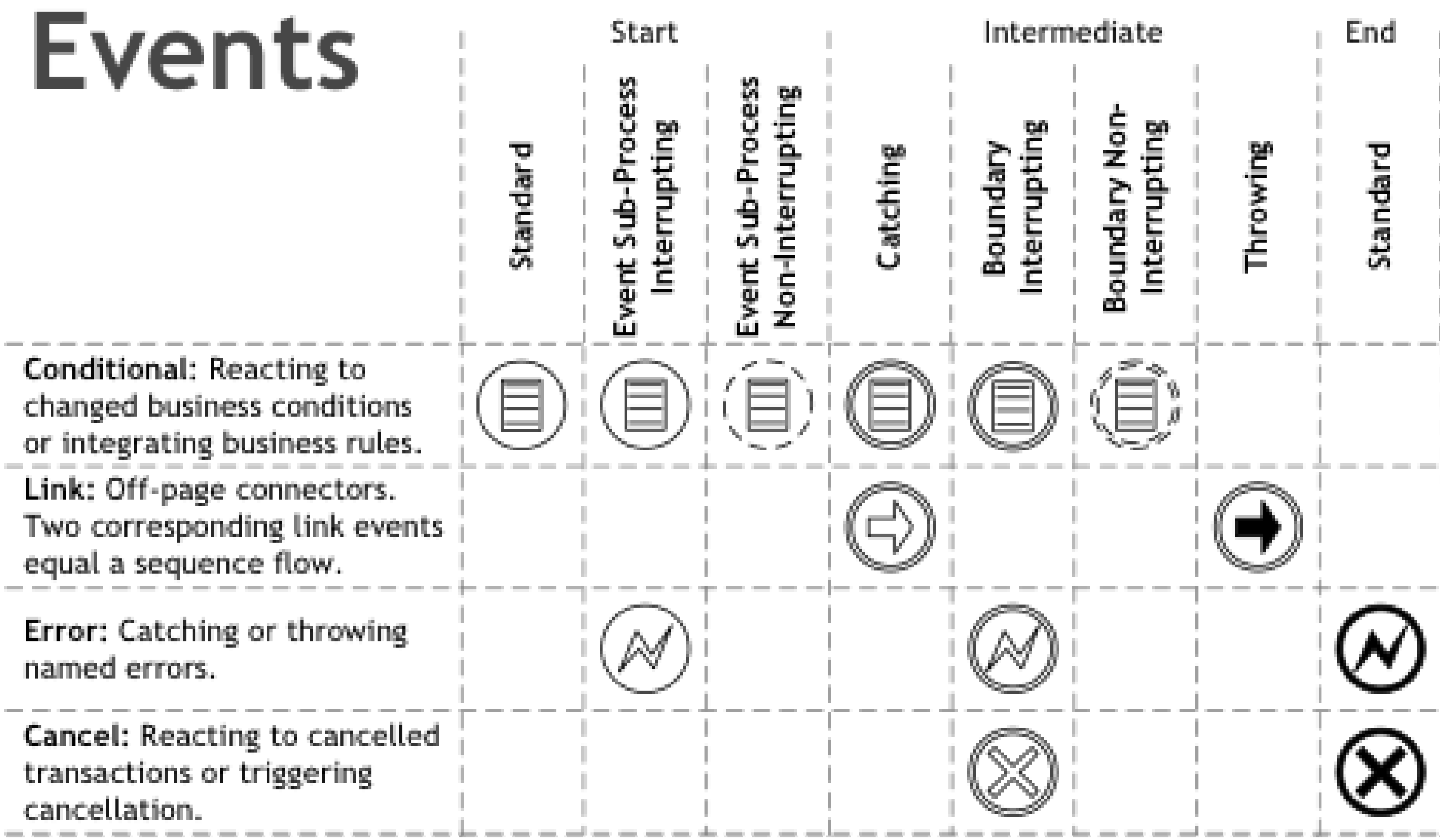

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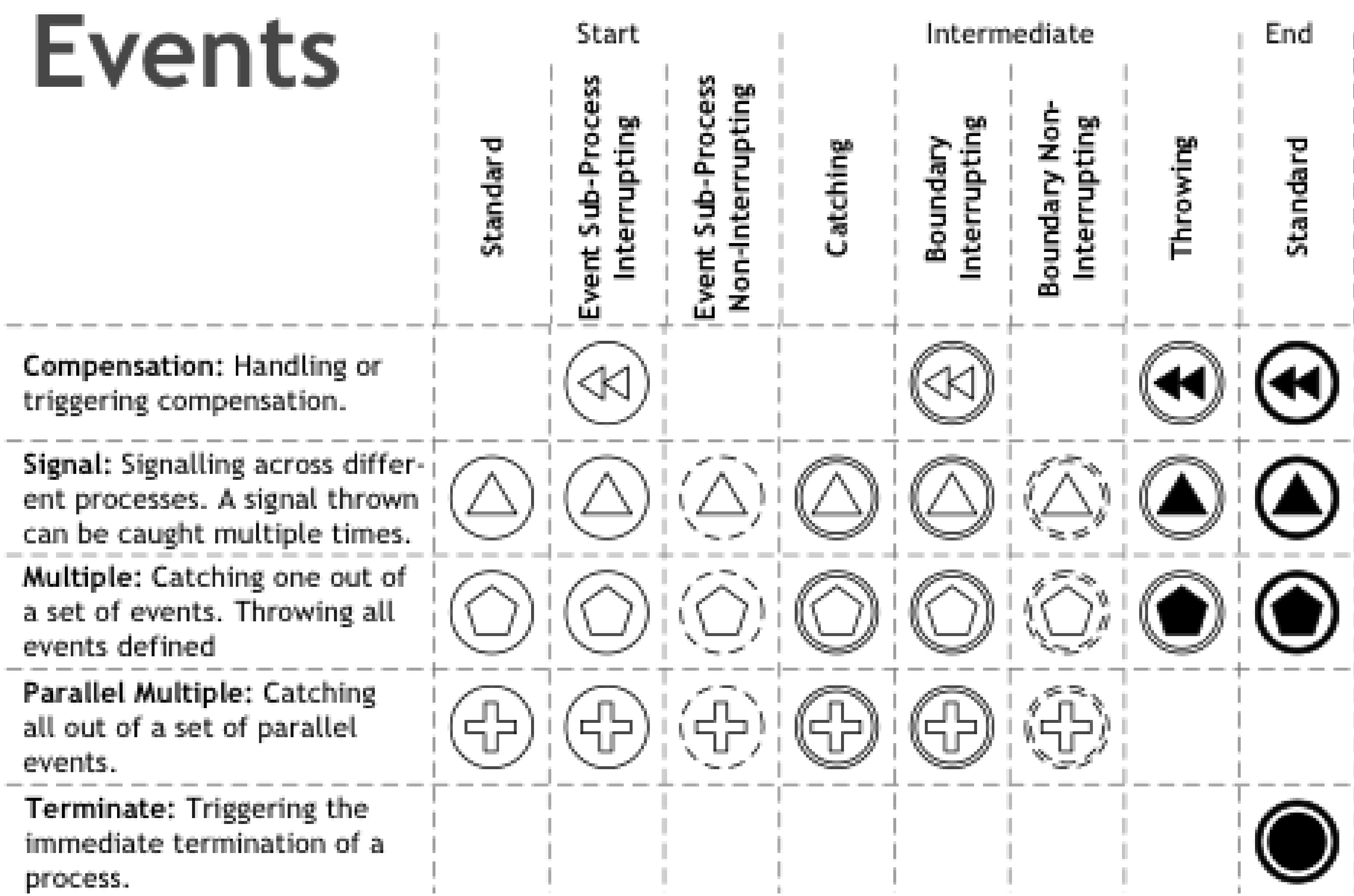
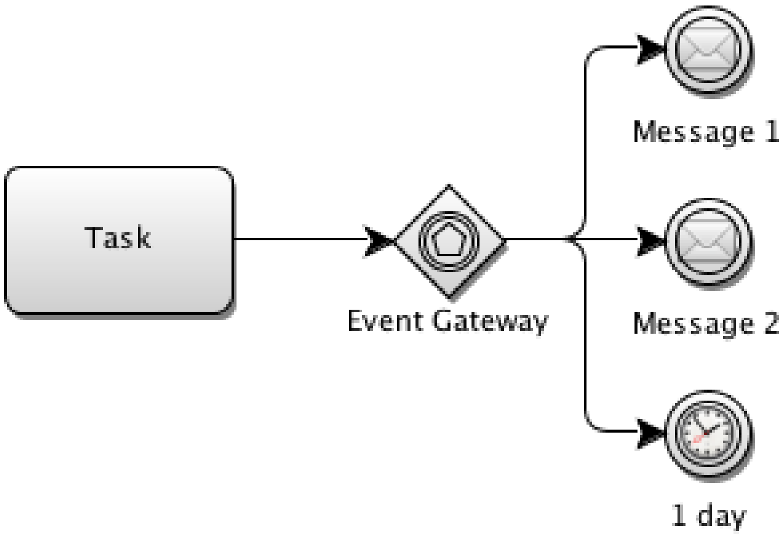


Event-Based Gateway

- Alternative paths are based on events that occur, rather than the evaluation of expressions
  - Usually the receipt of a message determines the path that will be taken
- When the first event is triggered, then the corresponding path is activated
  - All remaining paths will no longer be valid
  - The event gateway is thus a race condition where the first event that is triggered wins
- Event gateways can be used to start the process according to the event occurred

Event-Based Gateway



Demo - Shipment Process of a Hardware Retailer

- When goods are ready to be sent, the warehouse worker starts packaging the goods and a clerk decides if this is a normal postal or a special shipment.
- If a special shipment is required, the clerk selects a carrier and prepares the paperwork.
- Otherwise a normal post shipment is fine and in this case the clerk checks if an extra insurance is necessary.
- If the extra insurance is required, the logistics manager prepares that insurance. In any case, the clerk has to fill in a postal label for the shipment.

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Participants

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Activities

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